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Incorporating Large Language Models in Automated Real Estate Valuation

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ABSTRACT

This article explores the potential of automated valuation models (AVMs) by incorporating features extracted by large language models (LLMs). Utilizing a dataset containing 9,842 residential housing transactions from Oslo, Norway, we extract quantitative features from residential property advertisements with GPT models. We integrate these nonstructured features into an XGBoost model alongside structured data from the same advertisements. The findings reveal that LLM-extracted features significantly boost AVM performance by improving root mean squared error (RMSE) and mean absolute percentage error (MAPE) by 24.31 and 15.25%, respectively. Notably, housing standard and parking are the most influential features. The more accurate property valuations by enhanced AVMs not only have practical implications for tax authorities, banks, real estate agents, and algorithmic traders but also contribute to the research on artificial intelligence.

KEYWORDS

Automated valuation model (AVM); large language models (LLM); OpenAI GPT models; real estate valuation; text recognition

JEL CLASSIFICATIONS

C55; R21; R31

1. Introduction

In this article, we provide a new approach that combines textual features from real estate advertisements with structured data in the automated real estate valuation. Automated valuation models (AVMs), alternatively referred to as algorithm valuation, real-time valuation, mass valuation, or computer-assisted mass appraisal (CAMA) models, are becoming increasingly prevalent and are used by tax authorities, banks, real estate agents, and algorithmic traders, among others (see, e.g., Oust & Steininger, 2024). Nonlinear AVMs in particular have proven to provide good valuation estimates compared to more traditional techniques.¹ These models capture a broader range of relationships, utilize data more precisely, and address the problem of multicollinearity better. In addition, explorative artificial intelligence (AI) is superior in visualizing certain effects.²

However, as these models are increasingly used in policy-relevant settings such as setting property taxes or informing housing policy, concerns have emerged about their

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ability to deliver accurate and equitable valuations in geographically heterogeneous markets, where housing characteristics and pricing dynamics vary significantly across regions (Reite and Oust 2023; Reite, 2023). This concern reflects broader regulatory priorities around valuation quality, model risk, and financial stability in real estate-backed lending (European Banking Authority 2025). A broader application of AVMs in the banking sector encompasses activities such as the valuation of loan commitments (European Banking Authority, 2021). According to Xiong et al. (2021), AVMs are used in 30–70% of mortgage origination processes. In doing so, Reite (2023) finds that foreclosure discounts are lower for properties for which the bank employs an AVM, but this requires that a reliable AVM exists for the property.

Housing advertisements consist of both structured and unstructured data, whereas the regression algorithms used in AVMs require numerical or categorical inputs. Thus, previous AVMs used to estimate property values are based on housing attributes that are already available in a structured format, such as total area, build year, and number of rooms. However, housing advertisements also include listing texts that contain a considerable amount of unstructured data relevant to the selling price, such as listing text and image captions. This includes narrative descriptions of dwelling standard, parking conditions, or recent renovations. The recent emergence of effective and reasonable access to large language models (LLMs) enables analysis, extraction, and quantification of important information contained within such listing texts, thus enhancing the dataset of available structured data for the AVM. We investigate whether and how AVM performance can be enhanced by incorporating LLMs for feature extraction derived from OpenAI's GPT models. We also explore how the inclusion of information extracted with help from LLMs into an AVM can increase our understanding of the perceived value of different dwelling features and develop a GPT model-based method for extracting quantitative data from housing advertisements. Rather than presenting a broad methodological innovation, this study provides a focused, modular example of how LLMs can be operationally integrated into an existing AVM framework. Using a narrowly defined yet high-quality dataset of Oslo housing transactions over a short time period, we demonstrate a replicable and transparent pipeline that tests the practical feasibility of LLM-enhanced valuation models.

Most AVM studies have focused on integrating various machine learning (ML) methods to enhance predictability and to compare these methods with traditional residential property valuation approaches.³ However, AVMs have also been used for the valuation of curb appeal and architectural styles (see, e.g., Johnson et al., 2020; Lindenthal & Johnson, 2021; Wan & Lindenthal, 2023), assessment of the property's technical condition on its price (Oust et al., 2023), evaluation of a property's interior (Despotovic et al., 2023), and feature extraction from floor plan image segmentation (Kringeland et al., 2025).

The use of textual information to construct variables for real estate valuation is not new. For instance, Haag et al. (2000) employed keywords from property listings in a hedonic pricing model. Building on this, Nowak and Smith (2017) incorporated textual data into a hedonic framework and reported a reduction in pricing errors of more than 25%. Shen and Ross (2021) further examined the role of textual property descriptions, demonstrating that uniqueness influences price levels and, importantly, that this effect

also persists within a repeat-sales framework. More recently, Oust et al. (2020) and Nowak et al. (2021) applied textual descriptions in an AVM context, showing that such data can improve real estate valuation accuracy. These studies clearly demonstrate the potential of extracting information from listing texts for use in real estate valuation. However, the methods they employ are considerably more demanding, requiring greater knowledge of coding and substantial manual effort even to capture relatively simple textual patterns, especially when compared to the opportunities provided by new LLM-based tools. There have been some limited attempts at more complex feature extraction from the textual part of housing advertisements, both with (Cetinoglu, 2023; Foster et al., 2013) and without the use of LLMs (Galtier, 2022). Du et al. (2025) and Geerts et al. (2025) also make use of LLMs, with Du et al. (2025) employing them to provide explainability for AVM models, while Geerts et al. (2025) explore the potential of LLMs as practical tools for real estate valuation, showing that they can generate competitive and explainable price estimates. To the best of our knowledge, however, no studies exist that combine the features extracted from listing texts with the structured data about the housing unit for price estimation to the same extent as is done in this work.

The NLP technology has significantly evolved since it was first introduced,⁴ with modern LLMs such as BERT and GPT demonstrating unique and powerful abilities for both text understanding and generation. Prompt engineering—formulating and designing the prompt in a way that will result in the desired output—has proven important for getting precise and high-quality responses from GPT models (Liu et al., 2023). The prompt guides the model more effectively and is a good alternative for fine-tuning models by unlocking the potential in GPT models for a certain task (OpenAI, 2024).

In our empirical analysis, we use a dataset containing 9,842 individual transactions of residential housing as well as advertisements on dwellings sold between January 2022 and March 2023 in Oslo, Norway. First, we develop a pipeline that accesses OpenAI's GPT models, takes housing advertisements as input, and returns quantitative values. Then, we perform prompt engineering based on the available literature to obtain quality data from the LLM and collect data from the pipeline. The LLM extracts features about parking, kitchen age, bathroom age, fireplaces, storage units, storage unit size, outdoor area, outdoor area size, and housing standard. After validating the effectiveness and reliability of the LLM, we develop an AVM using both the structured data from the advertisements and the unstructured data from the LLM. Because gradient boosting machines have shown high accuracy in predicting real estate prices (see Birkeland et al., 2021; Chen et al., 2023; Helgaker et al., 2023; Hjort et al., 2022; Kok et al., 2017; Kumkar et al., 2018; Pollestad et al., 2024), we use XGBoost as our preferred AVM. Finally, we validate the effectiveness and reliability of the AVM.

We find that LLMs can be used to improve AVM performance. Compared to the model without the features extracted by the LLM, the XGBoost valuation model root mean squared error (RMSE) was reduced from 10.53 to 7.97% and the mean absolute percentage error (MAPE) was reduced from 6.36 to 5.39%. The features with the most positive impact were housing standard, fireplace, and parking. Our findings can be used to improve the AVMs utilized by tax authorities, banks, real estate agents, and algorithmic traders, as well as for enhancing AVMs used for research purposes. Our study also demonstrates that LLMs

can be used to enrich structured housing data with text-derived features, and it provides a clear and replicable pipeline for doing so

The remainder of this article is structured as follows: [Section 2](#) provides a brief introduction to the Norwegian housing market and describes our data gathering and filtering process. [Section 3](#) outlines the methodologies on feature extraction by LLM and property valuation by AVM. [Section 4](#) presents the results of both the LLM and AVM evaluation. We offer concluding remarks in [Section 5](#).

2. Background and Data

2.1. Norwegian Housing Market

In the Norwegian market, the most common method of selling residential property is through an open English auction. Over 98% of all residential transactions are conducted through a real estate agent (Flått et al., 2022). The seller employs a real estate agent who acts as both an independent third party and an auctioneer (Sønstebo et al., 2021). This characteristic, coupled with the fact that nonprofessional buyers and sellers dominate the Norwegian housing market, makes this context highly suitable for studies on property pricing in general (Oust et al., 2020).

Our data relate to the property sales in Oslo, the capital of Norway, which is divided into 17 districts. As of December 2023, the city had around 720,000 inhabitants. Over the past few decades, the city has experienced significant and steady growth; approximately half of Norway's population growth during this period has occurred in Oslo. Historically, the eastern part of Oslo has had generally smaller and cheaper dwellings compared to the western part (Birkeland et al., 2021).

2.2. Data Description

The dataset used in this study contains 9,842 residential housing advertisements from finn.no, the main platform for listing real estate for sale in Norway. The included dwellings were sold between January 2022 and March 2023 in Oslo, Norway. The dataset was provided by Solgt.no, a Norwegian Proptech company, and consists of traditional structured data about the dwellings sold as well as the advertisement text. [Figure 1](#) presents a heat map of Oslo that visualizes the frequency of property transactions included in our dataset. The majority of transactions have taken place in the city center and toward the east of Oslo.

The traditional structured dataset is comprised of 25 features, including address, sale date, and sale price. It also contains quantitative data, including size, number of floors, rooms, bedrooms, bathrooms, and the year in which the building was constructed. Additionally, the dataset includes information on ownership type, housing type, and geographical features like district, longitude, and latitude. The dataset with the advertisement text contains all textual information from the housing advertisements, i.e., title, description, and image captions, which we denote as “listing text.” The listing text data is composed of clean and complete sentences from the advertisements, with unnecessary whitespaces removed. [Table 1](#) provides summary statistics for each of the 24 usable features in the dataset (the “street address” feature was not used) including information

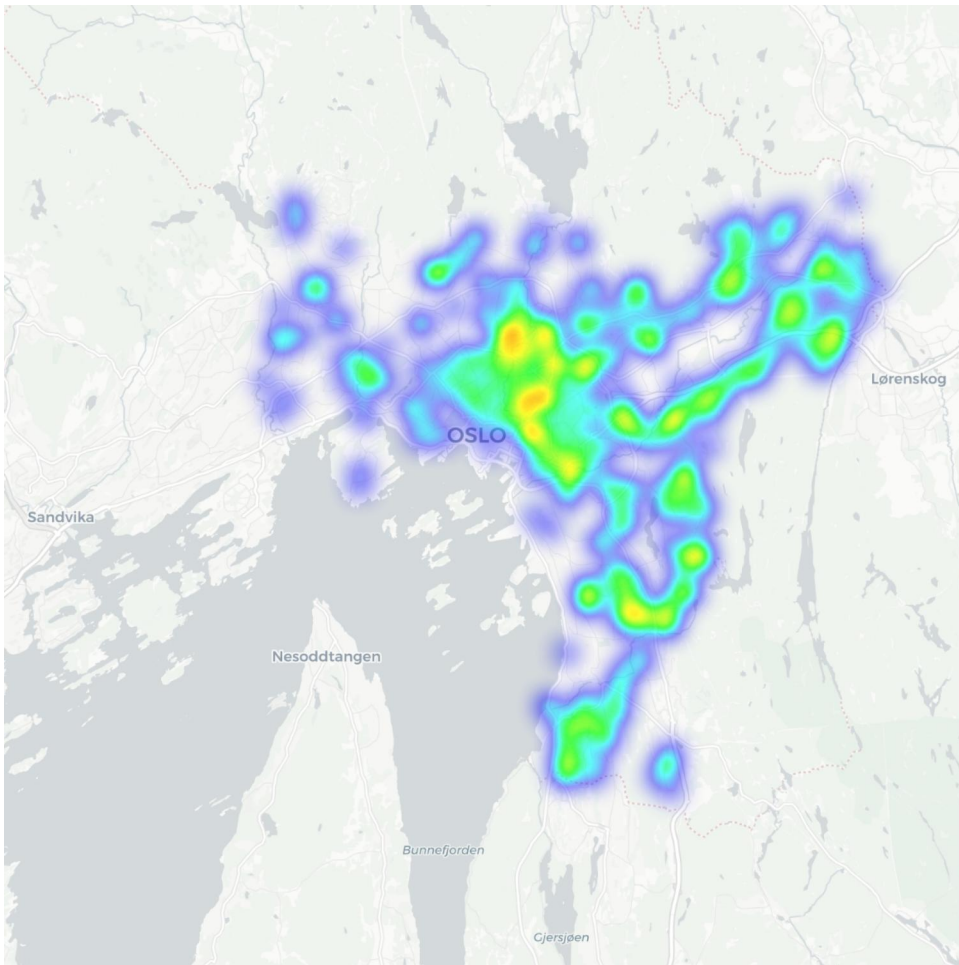


Figure 1. Heat map of residential property transactions.

This figure shows the heat map of the residential property transactions included in the dataset, superimposed on the map of Oslo and its surroundings. A warmer color indicates a higher number of transactions in the area.

about the advertised dwelling, such as size, number of rooms, bathrooms, and housing type. Additionally, it provides details about the area and location of the property, including the postal code, district, longitude, and latitude.

3. Methodology

3.1. Research Design

We begin with housing advertisements sourced from finn.no, which include both structured and unstructured information. In a first step, we use an LLM for feature extraction. In a second step, the feature values generated by the LLM are integrated into the AVM alongside the structured information to test the impact of the constructed LLM features. [Figure 2](#) provides a visualization of our research design.

Table 1. Summary statistics on features in the dataset.

Feature	Unit	Min	Mean	Max	Type
Price	MNOK	1.65	4.95	19.35	Numerical
Common debt	MNOK	0	0.30	5.94	Numerical
Common cost	NOK	530	4,557	21,490	Numerical
Housing type	–	–	–	–	Categorical
Ownership type	–	–	–	–	Categorical
Building type	–	–	–	–	Categorical
Energy label	–	–	–	–	Categorical
Building year	–	1860	1958	2023	Numerical
BRA	m ²	2	65	365	Numerical
PROM	m ²	13	63	365	Numerical
Lot Size	m ²	0	105	4,142	Numerical
Floor	–	1	2.98	19	Numerical
Rooms	–	0	2.71	7	Numerical
Bedrooms	–	0	1.74	6	Numerical
Bathrooms	–	0	0.76	3	Numerical
WC	–	0	0.80	4	Numerical
Elevator	–	–	–	–	Binary
Balcony	–	–	–	–	Binary
Parking	–	–	–	–	Binary
Longitude	Degrees	–	–	–	Numerical
Latitude	Degrees	–	–	–	Numerical
Postal code	–	–	–	–	Numerical
District	–	–	–	–	Numerical
Date sold	–	–	–	–	Date

Note. This table shows the minimum, mean, and maximum value of the price including common debt given in million NOK (MNOK). Common debt and common cost are the common loans of the housing association and monthly common costs, respectively. Housing types include detached home, duplex, row house, apartment, and other. Ownership type covers housing cooperative, sole ownership, and other. Building type uses standard categories according to their primary use or function. This includes the land, size, and addresses. Energy label is denoted on a scale from A to G for the dwelling's expected energy consumption. BRA is the size of the total area in m². PROM is the size of the living area in m². The floor indicates the floor the dwelling is located on. For detached homes, the floor is set to 1. If a dwelling has multiple floors, the lowest floor is used. Rooms is total number of rooms in the dwelling. Bedrooms and bathrooms are the number of bedrooms and bathrooms in the dwelling, respectively. WC is the number of additional bathrooms in the dwelling. Parking indicates if there are parking facilities adjacent to the property. A standard four-digit postal code indicates a geographical area in Norway. District or basic statistical unit is a numerical ID for "grunnkrets," used for geographical areas for statistical and property purposes in Norway.

3.2. Feature Extraction with LLM

Housing advertisement websites contain a significant amount of unstructured information. Thus, in this first step, the aim is to create a program that leverages AI's NLP capabilities to convert this textual information into quantitative values by defining a set of rules for the model to follow for each feature. The program will take the entire listing text as input and will generate a score for each feature based on the predefined rules. The features selected for extraction by the LLM, along with the corresponding explanations, are listed in Table 2. Notably, these features were not included in the structured information available for the listed dwellings. The selection of these features was guided by their relevance in the pricing process of a dwelling.

LLM Program: The first part of the program reads the dataset containing housing advertisements. This dataset includes the title, image captions, and description, which are combined into the advertisement listing text.

The next part of the program involves accessing the GPT model and generating a response from it. Newer models utilize what is known as the Chat Completions API endpoint, which requires a specific format for defining the model input. To generate a response from the LLM, the input must be structured to align with the model's expected

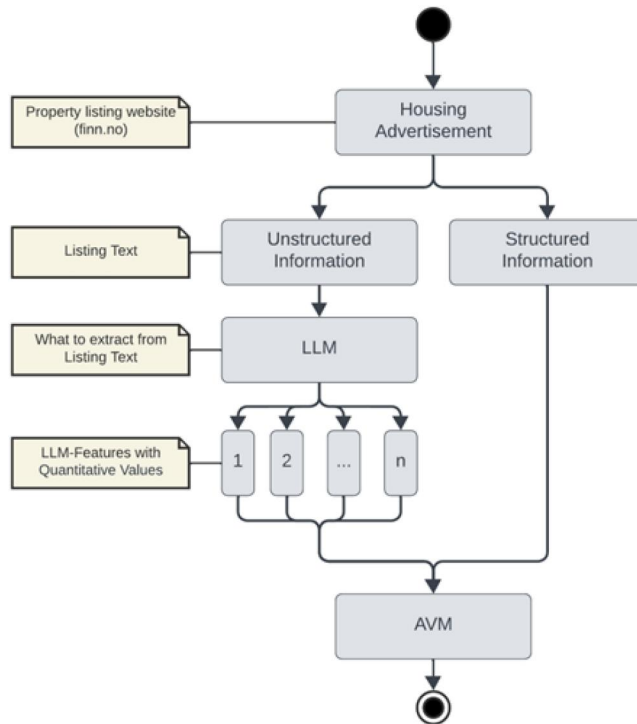


Figure 2. Visualization of the process.

The figure provides flowchart visualization how structured and unstructured information is extracted from the housing advertisement, how the unstructured information is processed, and how both the structured and unstructured information are then used in the automated valuation model.

Table 2. LLM features.

Feature	Explanation
Parking	If the unit has special parking rights
Kitchen	Year of the last upgrade
Bathroom	Year of the last upgrade
Fireplace	Number of fireplaces
Storage	Number of storage units
Storage size	Total size of storage units in m ²
Outdoor area	Type of outdoor area
Outdoor area size	Total size of outdoor area in m ²
Housing standard	Overall impression of the dwelling

Note. This table lists the features for LLM extraction. All features have numerical values.

behavior. This involves creating a prompt that instructs the model on how to analyze the listing text. The combination of the prompt and the model's instructions is referred to as a "message" and its definition is as follows:

```

message = [
  {"role": "system", "content": f"{model_behavior}"},
  {"role": "user", "content": f"{prompt}"},
]

```

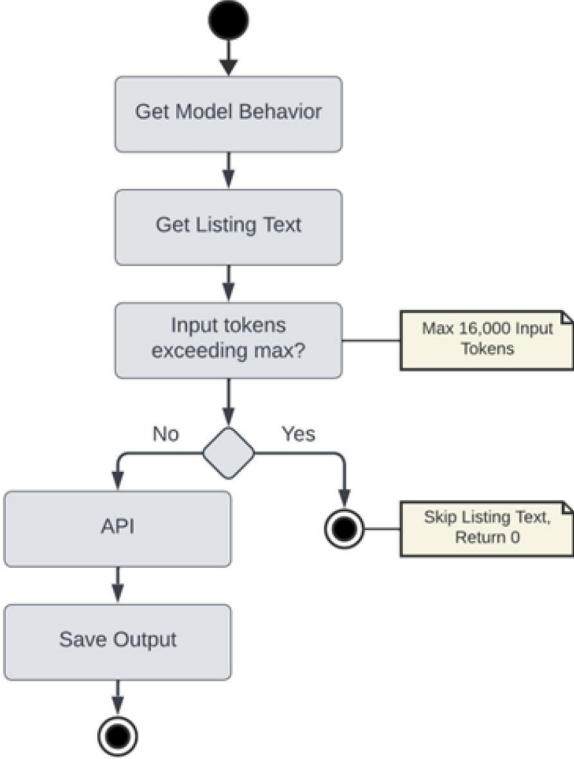



Figure 3. Visualization of preparing and creating message by LLM. The flowchart also shows how the program handles cases when a message exceeds the input token limit.

The model behavior is a string constructed as a set of rules for the model to follow for each feature. The full set of rules is detailed in the Appendix. This string specifies what the model can expect as input—in this case, a listing text—and how it should return the extracted information based on the rules for each feature. As the program accesses the GPT model once per cycle, one message is sent per listing text. The process of prompt engineering to achieve the desired output in the correct format is described below. Figure 3 illustrates the message creation process.

OpenAI offers several models with varying price ranges, limitations, and performance levels. We chose OpenAI’s GPT-3.5 model, specifically gpt-3.5-turbo-0125, which has a limit of 16,000 input tokens. Both the model behavior and the prompt count toward the upper limit for input tokens. However, the program is designed to handle cases where the total number of input tokens exceeds this limit. The LLM output is a string containing all features and their corresponding scores in the following format: ‘parking’: ‘p_score’, ‘kitchen’: ‘k_score’, ‘bathroom’: ‘b_score’, and so on for all features, where ‘x_score’ represents the most probable value returned by the LLM based on the model behavior.

Because in Norway property advertisements are typically written in Norwegian, the model behavior is also written in Norwegian. This ensures consistency between the model behavior and the prompt, making it easier to verify if the model has missed any information when reviewing the output results. The output from the LLM is a string

containing all features and their corresponding scores. The next part of the program processes this output string by cycling through the features and extracting the relevant scores. These scores are then systematically saved in a .CSV file, resulting in a set of new LLM-derived features for use by the AVM, as detailed in [Table 2](#).

Prompt Engineering: During program development, extensive prompt engineering was done to make the model behavior as precise as possible, following the recommendations from OpenAI (2024) for formulating inputs. The final version of the model behavior is based on observations from earlier drafts and is more direct and concise, resulting in a more consistent output format and values. An excerpt from the prompt used reads⁵:

Your task is to analyze the content of the real estate ad. You should return the following information about the property in the following format: (...) "kitchen": "k score," (...). "k score" is equal to the calendar year (in the YYYY format) the kitchen was built or last renovated, refurbished, or upgraded, or alternatively, the year the property was built. If the year is not mentioned, return 0.

Testing Process for the LLM: The effectiveness and reliability of the LLM was validated by conducting several tests and comparing the results to manually labeled data extracted from 200 housing advertisements based on the same rules as those described in the model behavior for the LLM. In total, 15 tests were conducted with the same model behavior and the same 200 housing advertisements. The tests were grouped into three sets, with five tests per set. This grouping allowed for direct comparison of the results both with the manually labeled data and among the different test groups. Each test involved a full run of the model based on the same set of 200 housing advertisements. For every run, the model generated predictions based on the rules described in the LLM's behavior. The overall goal of the tests was to evaluate how accurately the model could replicate the manually labeled values, as these served as the "correct" reference, providing insight into its overall performance and reliability. To evaluate performance, several validation methods were used, including precision-recall analysis, correlation analysis, and percentage-based accuracy against the manual dataset. Since the test conditions remained unchanged, any variations in results were attributed to natural fluctuations in the model's responses rather than differences in the test setup. This approach allowed for a thorough assessment of the model's stability and performance across multiple iterations.

3.3. Automated Valuation Model

In the second step, we test the constructed LLM variables by examining their influence on the housing prices.⁶ For this purpose, we interpret the LLM variables in various AVMs.

Data preprocessing: As our raw dataset has several missing values, some data preprocessing is required before running the AVMs. However, because our dataset is small, merely removing rows with missing values would negatively affect the model performance and would result in biased results. Thus, instead of removing the incomplete entries, we impute the missing values using estimation techniques based on the characteristics of the missing variables. As shown in [Table 3](#), nine of the available features have missing values, which are replaced by the variable's mode, mean, or median value. The standard structured variables were imputed using mode or mean, dependent on their distribution. The LLM is instructed to return 0, except for parking, if it is unable to

Table 3. Handling missing values (Not a Number [NaN]).

Feature	Number of NaN	NaN Replacement
Bedrooms	213	mode
District id	45	mode
Building type code	353	mode
Common Costs	28	mean
BRA	119	mean
Building year	12	mean
Lot size	861	mean
Kitchen	4,730	median
Bathroom	3,266	median

determine the appropriate value (see the Appendix for details). For the LLM features kitchen and bathroom, the missing values (the 0s) were imputed with the median. For the other LLM created features the 0 value was kept as the lack of information about this feature indicates it does not exist (e.g. for fireplace, storage size), or for the regressor to determine the value of the “information missing” category (housing standard).

The feature sold date was converted to sold date days, which transforms the data from a year-month-date format to numerical values representing the number of days since the property was sold relative to a reference day. Both housing type and ownership type are features with categorical values, for which we created dummy variables for each type. The housing type variable was divided into detached, duplex, row house, apartment, and other, while the ownership type was divided into sole ownership, co-op, and other. The features kitchen, bathroom, and building year all have values expressed as calendar year. These values were converted to age, directly providing information on how old the kitchen, bathroom, or building is. This transformation simplifies the model by eliminating the need to handle these relationships separately.

Finally, we used Isolation Forest to identify and remove outliers from the dataset. This method was chosen as it has been shown to be superior to other approaches, especially for datasets with many abnormal deviations (Liu et al., 2008). The Isolation Forest model identified 1,041 possible outliers, which is equivalent to 10.58% of the dataset. Given the sample size, removing too many data points could eliminate genuine values and reduce diversity in the data. Further testing showed that removing 9.5% of the possible outliers provided the best results while retaining most of the dataset. This process filtered out 1%, corresponding to 99 samples, reducing the original dataset from 9,842 to 9,743 property listings.

Linear Regression: We chose a hedonic ordinary least squares (OLS) estimator as our linear base model. Hedonic pricing models determine the value of an asset by breaking it down into its constituent parts or characteristics. They are frequently applied in property market analysis to estimate dwelling prices, based on the assumption that a dwelling’s price reflects the product of its components (Brooks, 2019). By analyzing individual components such as size, location, and number of rooms, we can assess how changes in one characteristic affect the overall price while holding other factors constant.

By employing the hedonic OLS model, our aim was to create a benchmark for our nonlinear model and provide broader recognizability. Since this is a linear model, certain variables need to be transformed, which would result in some loss of information compared to a nonlinear model. Categorical variables such as city districts, energy labels, housing types, and ownership types are converted into dummy variables. Similarly, the

Table 4. Hyperparameter tuning range.

Hyperparameter	Value Range
Learning rate	(0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9)
Gamma	(0.0, 0.1, 0.2, 0.3, 0.4, 0.5)
Min child weight	(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)
Subsample	(0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0)
Colsample bytree	(0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0)
Max depth	(3, 4, 5, 6, 7, 8, 10)
N estimators	(100, 200, 300, 400, 500, 600, ..., 1,800, 1,900, 2,000)
Alpha	(0, 0.01, 0.1, 1, 10, 100)
Lambda	(0, 0.01, 0.1, 1, 10, 100)

building year is divided into five groups, each containing 20% of the data. The variable for the time of sale is also divided in a comparable manner, with 25% of the dataset in each group. The objective of an OLS estimator is to determine an accurate relationship between the independent variables and the dependent variable as shown below:

$$\ln(P_{it}) = \alpha_0 + \delta_t + \sum_k \beta_k X_{kit} + \varepsilon_{it}, \quad (1)$$

where $\ln(P_{it})$ is the natural logarithm of the dwelling price per square meter, X is a vector of explanatory variables for the presence of certain characteristics, k , and δ_t are time dummies for the point in time when the dwelling is sold, t denotes the time period, and i refers to the specific dwelling. Lastly, ε_{it} is the error term for dwelling i at time t .

Among the LLM variables, there are two that are not naturally suited to being treated as linear. For this reason, we handled these LLM variables differently in the linear and the nonlinear models. These are parking and outdoor area, for which we created dummy variables. We also created dummy variables for housing type, ownership type, energy label, building year, time of sale, and location from the original dataset. Time of sale was constructed by dividing the dataset's time span into four periods/dummies, each covering approximately 15 months and containing an equal number of dwelling sales. For location, we used dummy variables for the 15 administrative districts in Oslo.

XGBoost: The powerful ML technique XGBoost is highly effective for regression tasks (Nielsen, 2016). XGBoost is typically one of the best-performing models in housing price prediction (Birkeland et al., 2021; Kok et al., 2017; Mayer et al., 2019). The main objective of the algorithm is to iteratively improve the predictive performance of a collection of weak learners, typically decision trees, by combining their outputs in a way that minimizes a predefined loss function (Chen & Guestrin, 2016). The algorithm is initially given a set of data, where x_i denotes features outlined in Table 1 (excluding address) and Table 2, and y_i is the target value represented by the dwelling price. The hyperparameter value ranges are provided in Table 4. The dataset was split into training (85%) and test (15%) subsets. Shapley additive explanations (SHAP) was our chosen explanatory AI (XAI) technique, as it assigns each feature a value based on its influence in the decision-making process and thereby quantifies the average marginal contribution of a feature.

4. Empirical Results

In this section, we present the results. First, we analyze the results from the GPT model and compare it with the manually labeled dataset, after which the testing and the AVM results are presented and analyzed.

Table 5. Average feature accuracy and correlation.

Feature	Accuracy	Average correlation	Correlation of the average
Parking	29.37%	0.7920	0.8186
Kitchen	92.87%	0.8730	0.8740
Bathroom	90.70%	0.8400	0.8450
Fireplace	95.47%	0.8455	0.8461
Storage	73.37%	0.6080	0.6171
Storage size	95.70%	0.8282	0.8547
Outdoor area	70.53%	0.7032	0.7463
Outdoor area size	93.57%	0.9801	0.9812
Housing standard	33.67%	0.7146	0.7474

Note. This table shows the average feature accuracy and the average correlation across the 15 tests compared to manually labeled data. The column Average correlation represents the average of the correlation coefficients computed individually for each of the 15 tests. In contrast, the column Correlation of the average refers to the correlation coefficient calculated based on the average values derived from the results of all 15 tests combined.

4.1. Feature Extraction with LLM

The average feature-wise model accuracy for all tests is summarized in Table 5, along with the correlation of each feature to the manually labeled data as an indication of how well the LLM results concur with the actual advertisement content. The output returned for all features is relatively consistent. The biggest difference between the results was 5.0% for housing standard, where the weakest result was 31.5% in Test 10 and the strongest was 36.5% in Test 11. The average difference between the weakest and strongest results for all features was 2.2%.

The model performs very well for most features, but the accuracy is significantly lower for parking and housing standard. This discrepancy can be partly attributed to the more complex rules adopted for these features when defining the model behavior. Despite the intricate rules and criteria, which may be challenging to apply perfectly, calculating the correlation coefficient helps to capture trends in the data. The coefficients for both features demonstrate relatively high correlations, indicating a stronger performance than initially suggested by the percent-wise accuracy figures. Compared to manually labeled data, the weakest and strongest tests for parking results in correlation coefficients of 0.7897 and 0.8085, respectively. For housing standard, the correlation coefficients for the plots were 0.7163 and 0.7235, also reflecting a stronger relationship and performance than the percent-wise accuracy initially suggested.

Given that language models have some inherent uncertainty and unpredictability in their responses, the correlations and results may vary. Table 5 summarizes the average correlation coefficients for parking and housing standard across separate groups. Observations from the tests suggest that a weaker (stronger) performance in one feature often correlated with a weaker (stronger) performance in the other. The average correlation across all features is higher than the average correlation for individual features, indicating better performance when averaging the LLM output.

In cases where one observation is significantly more frequent than others, the precision-recall method is a useful way to validate model results (Fränti & Mariescu-Istodor, 2023). This is particularly relevant for some dwelling features. For example, in our case, 77% of the listed dwellings did not have a fireplace. The precision-recall method shows a trade-off in which high recall and low precision mostly yield incorrect predictions. Recall measures the number of relevant results returned, while precision

Table 6. Summary statistics on LLM features in the dataset.

Feature	Unit	Min	Mean	Max	Type
LLM Parking	–	0	3.97	7	Numerical (categorical)
LLM Kitchen age	Year	1	9.76	224	Numerical
LLM Bathroom age	Year	0	10.74	87	Numerical
LLM Fireplace	–	0	0.27	3	Numerical
LLM Storage	–	0	1.41	7	Numerical
LLM Storage size	Sq.m	0	2.38	91	Numerical
LLM Outdoor area	–	0	5.83	10	Numerical cCategorical)
LLM Outdoor area size	Sq.m	0	6.90	450	Numerical
LLM Housing standard	–	0	4.85	8	Numerical (categorical)

Note. This table shows the minimum, mean, and maximum value of the LLM variables. For LLM kitchen age and LLM bathroom age, the age is calculated based on the difference between the year the room was renovated and the year of sale. Sq.m is an abbreviation for square meter.

Table 7. Model performance.

Model	LLM Features	RMSE	MAPE	Improvement RMSE	Improvement MAPE
Linear	No	17.65%	9.26%		
Linear with LLM	Yes	16.42%	8.51%	6.96%	8.10%
XGBoost	No	10.53%	6.36%		
XGBoost with LLM	Yes	7.97%	5.39%	24.31%	15.25%

Note. This table shows the best-performing model for each metric highlighted in bold.

measures the relevance of those results. Nonetheless, 14 out of 15 tests produced identical predictions for fireplaces, demonstrating both consistency and strong results, with an *F*-score of 0.9072, precision (*P*) of 0.8627, and recall (*R*) of 0.9565. Test 12 deviated slightly, with a lower precision of 0.8462, resulting in an *F*-score of 0.8980. These results indicate that the GPT model performs well. Overall, the LLM shows good accuracy and correlation for all features when compared to manually labeled data. This finding indicates that the GPT model can extract significant structured data from the listing text, enhancing the dataset quality and utility.

Observations from these tests, along with smaller initial experiments, emphasized the importance of well-structured prompts and clear rules for the LLM. The early trials laid the foundation for the prompt engineering process, while the results from these tests reinforced and validated these initial findings.

Table 6 provides summary statistics for each of the 9 LLM features extracted from the advertisement texts using GPT models.

4.2. Automated Valuation Model

In this section, we discuss the use of four AVMs: a linear OLS model without LLM features, a linear model with LLM features, and similarly, a nonlinear XGBoost model without LLM features and a nonlinear model with LLM features. First, we present the performance of the horse race among the four models. We then will interpret the features.

Model performance: Table 7 presents a comparison of the linear OLS models with and without the LLM features, and similarly the nonlinear XGBoost model with and

without the LLM features. As expected, we find that the XGBoost models perform significantly better than the linear models, given that the nonlinear models are better able to utilize the information related to important variables, such as location, sale date, and year of construction. More importantly, the LLM features lead to a significant improvement in performance for both linear and nonlinear models. For the linear OLS model, the RMSE declines from 17.65 to 16.42%, and the MAPE decreases from 9.26 to 8.10%, representing improvements of 6.96 and 8.44%, respectively. For the nonlinear XGBoost model, the RMSE declines from 10.53 to 7.97%, and the MAPE decreases from 6.36 to 5.39%, representing improvements of 24.31 and 15.25%, respectively. The significantly greater improvement seen for the XGBoost model, compared to that of the OLS model, indicates that the price effect of the LLM features is nonlinear (e.g. the relative price increase of having a parking space can be higher for a small apartment than for a house).

While all the models demonstrate strong performance, the XGBoost model with LLM features has exceptionally low RMSE and MAPE values. Additionally, the substantial improvement achieved by including the new LLM features suggests that these variables contain important information for property valuation. This highlights the significant value of the XGBoost model in interpreting the LLM features.

Model Interpretations: We first briefly examine the coefficients of the LLM features in the OLS model, before focusing on the XGBoost model. We then explore the LLM features further by comparing the SHAP Dependence Plots to the actual housing prices.

Table 8 shows the linear OLS regression results. The age of the kitchen and bathroom, storage size, number of storage units, fireplace, and housing standard are treated as linear variables, while dummies were created for parking and outdoor area. While most of the results are expected, and will be presented in more detail in connection with the XGBoost model, some exceptions are noteworthy. For example, *Parking Missing_{LLM}* is the parking dummy with the most positive coefficient. There are very few units for which the model was unable to return a value about parking. As street parking is the reference group, it is also possible that this information is not given for homes where private parking is expected, such as single-family houses. Furthermore, one might have expected the coefficient for *Kitchen Age_{LLM}* to be significantly negative. These results will be studied further in the SHAP analysis of the nonlinear XGBoost model. The coefficient for *Outdoor Area Terrace Shared_{LLM}* will also become more interpretable when we look at the SHAP analysis.

Figure 4 visualizes the feature importance for the XGBoost model with LLM features. We see that several of the LLM features are included among the 25 most important features. We recognize that variables such as size, location, and housing type have the greatest impact on price, while the new LLM variables also carry significant importance. Housing standard is among the features with highest importance, along with fireplace, parking, outdoor area, outdoor area size, and kitchen age.

Figure 5 shows the SHAP values for the LLM features in the XGBoost model. The clustering of blue dots on the left and red dots on the right indicates a positive relationship. As expected, the plot shows a clear positive relationship for *Housing Standard_{LLM}*, *Fireplace_{LLM}*, *Parking_{LLM}*, and *Outdoor Area Size_{LLM}*. For *Bathroom Age_{LLM}* and *Kitchen Age_{LLM}*, the relationship is negative, which is also expected. However, for *Outdoor Area_{LLM}*, *Storage Size_{LLM}*, and *Storage_{LLM}*, the relationships are less clear.

Table 8. Linear regression.

LLM features	OLS without LLM Features		OLS with LLM Features	
	Coeff.	Std. Err.	Coeff.	Std. Err.
Parking Missing _{LLM}			0.089***	0.033
Parking None _{LLM}			0.010	0.009
Parking Shared _{LLM}			0.010	0.010
Parking Rented _{LLM}			0.008*	0.005
Parking Seniority _{LLM}			0.008	0.006
Parking Private _{LLM}			0.024***	0.004
Parking Garage _{LLM}			0.027***	0.005
Kitchen Age _{LLM}			−0.00005	0.00005
Bathroom Age _{LLM}			−0.001***	0.000
Fireplace _{LLM}			0.041***	0.003
Storage _{LLM}			0.001	0.001
Storage Size _{LLM}			0.002***	0.000
Outdoor Area Backyard Private _{LLM}			0.023**	0.009
Outdoor Area Backyard Shared _{LLM}			−0.009	0.007
Outdoor Area Garden Private _{LLM}			0.065***	0.008
Outdoor Area Garden Shared _{LLM}			0.020*	0.012
Outdoor_area_Terrace Private _{LLM}			0.015	0.011
Outdoor Area Terrace Shared _{LLM}			0.024***	0.007
Outdoor Area Balcony Private _{LLM}			0.025***	0.007
Outdoor Area Balcony Shared _{LLM}			0.012	0.017
Outdoor Area Roof Terrace Private _{LLM}			0.024***	0.008
Outdoor Area Roof Terrace Shared _{LLM}			0.004	0.014
Outdoor Area Size _{LLM}			0.0003**	0.000
Housing Standard _{LLM}			0.033***	0.001
Constant	49.776***	8.778	49.776***	8.778
Dwelling attributes	Yes		Yes	
Time FE	Yes		Yes	
Location FE	Yes		Yes	
Adjusted R^2	0.816		0.842	
Number of Observations	9,842		9,842	

Note. This table shows dwelling attributes: common debt, common cost, dummies for housing type, dummies for ownership type, dummies for energy label, dummies for building year (five groups with approximately 2,000 dwellings in each), size (BRA, PROM), lot size, floor number, number of bathrooms, number of rooms, latitude, longitude, number of bedrooms, number of wcs, dummy elevator, dummy balcony, dummy parking; time FE: the 15 months are divided in to four periods/dummies with equal numbers of dwelling sales in each; Location FE: Dummies for city districts. Reference groups for LLM features: street parking, outdoor area not mentioned. ***, **, and * indicate significance at 1%, 5%, and 10% level, respectively.

To investigate the LLM features further, their performances are presented as SHAP dependence plots and are compared to the actual housing prices in Panels A–F of Figure 6. The value of the feature is displayed on the horizontal axis, while the vertical axis shows how much the value of a feature changes the final model prediction. Each dot representing an individual dwelling in the dataset, and the bars at the bottom of each chart represent histograms illustrating the distribution of the feature.

In Panel A of Figure 6, we observe a clear positive connection between housing standard and price. The description of the scale can be found in the Appendix. The pattern is most evident for scores 2 to 7, where most of the observations are clustered, and is somewhat less clear for scores 1, 8, and 9, where there are few or no observations. Score 0 = “Not possible to determine the standard” appears to be linked to higher prices than scores 1 and 2, but is lower than the average, as expected; however, these relationships are based on very few observations. This result, combined with the fact that housing standard is a variable that one would intuitively consider important—as confirmed by Oust et al. (2023)⁷—along with the traditional difficulty of gathering reliable data on this

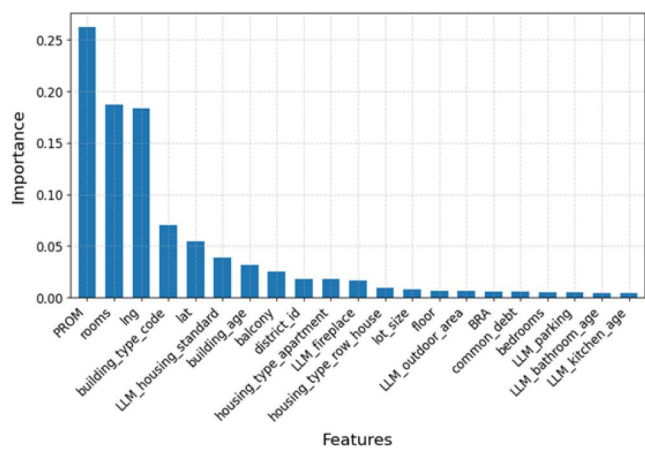


Figure 4. Feature importance of XGBoost model with LLM features. This figure illustrates the relative importance of the attributes, with larger bars indicating higher importance. The plot only displays 25 of the most important features due to the lack of space.

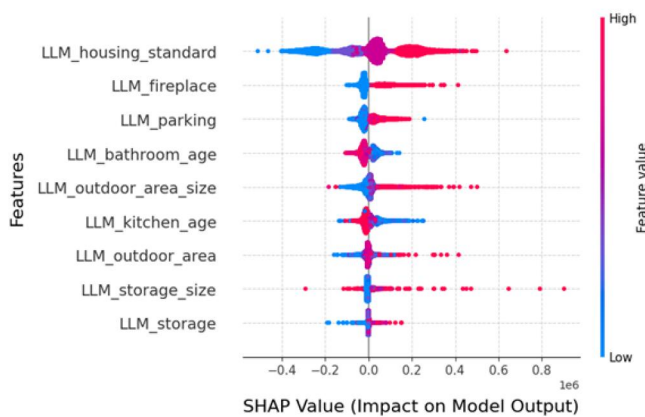


Figure 5. SHAP values for the LLM features in the XGBoost model. This figure illustrates the identified relationship between price and the LLM features. The SHAP value represents the estimated impact on the square meter price in NOK. For each attribute, the dots represent individual dwellings in the dataset. Clustering of blue dots to the left and red dots to the right indicates a positive relationship, i.e., a better condition results in a higher price.

characteristic, suggests that LLM and OpenAI’s GPT models could be a good way to obtain this property information.

Panel B of **Figure 6** shows that the *Parking_{LLM}* feature exhibits a U-shaped relationship with price (with a rather flat bottom). It is important to remember that the scale is not designed in a way that one should expect a linear or even increasing relationship to do as the numbers rise. The *P*-scores 2 – 5 of *Parking_{LLM}* includes all parking options where the property does not come with a designated parking space but where various parking possibilities are mentioned. At the same time, we see that *P*-score values 6 (has its own outdoor parking) and 7 (has its own garage parking) have a clearly positive impact on

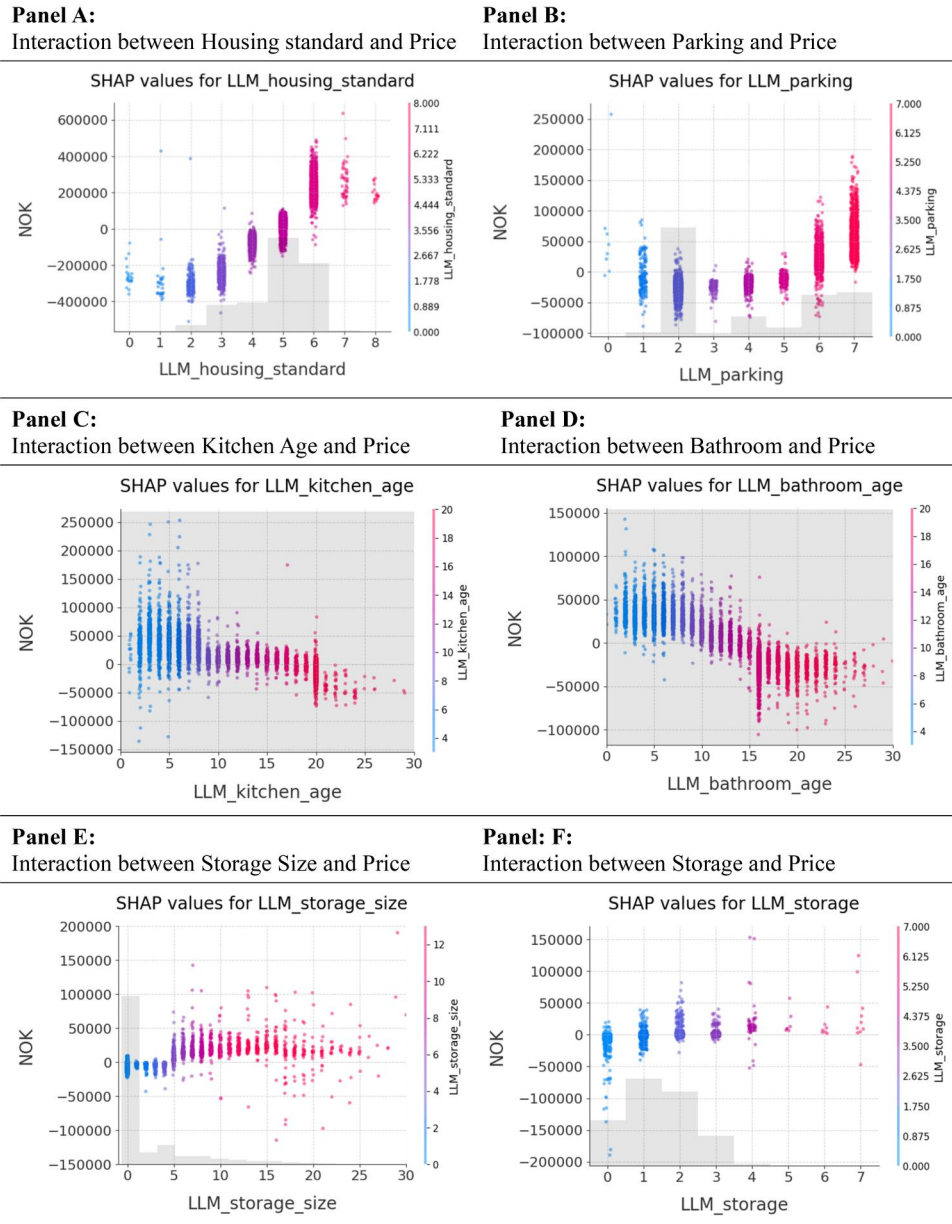


Figure 6. SHAP dependence plot.

This figure shows SHAP dependence plots for interaction between housing standard, parking, kitchen age, bathroom age, outdoor area, outdoor area size, fireplace, and price in Panels A–I, respectively. The value of the feature is displayed on the horizontal axis, while the vertical axis shows how much the value of a feature changes the final model prediction. The descriptions of scales for *Housing Standard_{LLM}* and *Parking_{LLM}*, *Kitchen Age_{LLM}*, *Bathroom Age_{LLM}*, *Storage Size_{LLM}*, *Storage_{LLM}*, *Outdoor Area_{LLM}*, *Outdoor Area Size_{LLM}*, and *Fireplace_{LLM}* can be found in the Appendix. 0 indicates that no value was returned by the model. The dots represent individual dwellings in the dataset, and colors represent the scale. The bars at the bottom of each chart represent histograms showing the distribution of the feature.

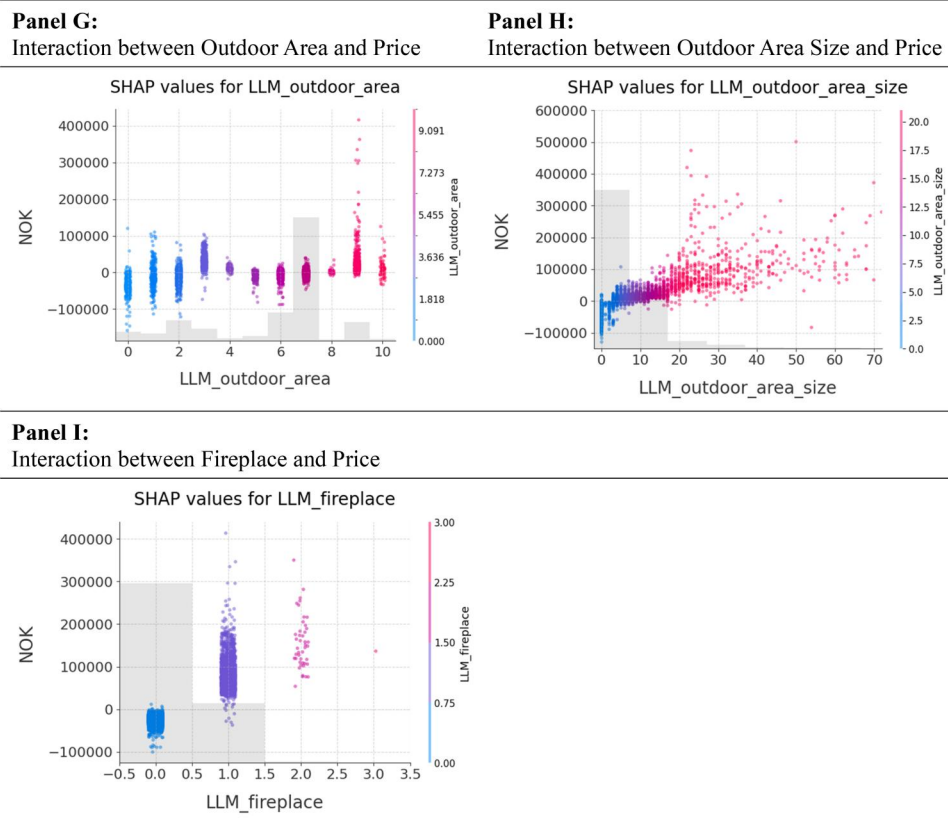


Figure 6. Continued

the price, which can be considered a useful contribution from the LLM. For P -score 1, where parking is not mentioned, there is a wide variation in how this feature influences the price. This is not surprising, since it is the actual parking option that affects the price, not what is stated in the sales listing. However, it is surprising that P -score 1 in some cases has a positive price impact, suggesting that for some properties with designated parking (corresponding to P -score 6 or 7), this information is omitted from the listing. This indicates that, despite the high interpretability of the results, a connection to economic reasoning is still required when interpreting them.

Panels C and D of [Figure 6](#) show *Kitchen Age_{LLM}* and *Bathroom Age_{LLM}*. It is evident that the longer it has been since the kitchen or bathroom was renovated, the lower the standard and therefore the lower the price. Here, the scale along the x-axis represents the number of years since renovation. For kitchens, we find a relatively flat price influence from 8 to 16 years, where the price impact varies around zero. If the kitchen is neither new nor old, the age has less significance. For bathrooms, this effect is not as clear. Although the number of years since the renovation of both the kitchen and bathroom has a clear negative correlation with price, the dots representing home sales show quite a wide variation. Moreover, for some dwellings where these rooms have recently been renovated, the model suggests a negative price impact. These findings

are similar to those reported by Oust et al. (2023), who used condition grades for these rooms, and it is likely that the LLM has successfully extracted accurate and valuable information.

Panel E of Figure 6 depicts *Storage Size_{LLM}*. It indicates that most properties have an SS-Score of 0 (corresponding to the “the size is not clearly stated” option) and that the price effect in these cases is close to zero, with surprisingly little variation. Furthermore, having a very small storage space has a negative impact on the price. In contrast, we see a positive price effect from larger storage size up to 10 square meters. If the storage size increases further, the price relationship becomes more unclear with a very wide variation. Panel F of Figure 6 shows the interaction between the number of storage units and price. Although there are few dwellings with more than three storage units, having no storage space appears to lower the property price, while having one storage unit seems to neither increase nor decrease the price. Having two storage units slightly increases the price further. However, having more than two storage units does not appear to result in a higher property price compared to having two storage units. Both variables describing storage make strong economic sense: having no storage or very limited storage entails a negative economic value, whereas beyond a certain point it may be more beneficial to allocate the square meters to other uses. Similarly, having two storage units appears to be more valuable than having one—for example, they may serve different purposes—but at some stage the marginal value of additional units diminishes.

Panel G of Figure 6 shows the interaction between the type of *Outdoor Area_{LLM}* and price. The variable values are explained in the Appendix. A value of 0 indicates that an outdoor area is not mentioned. Odd-numbered values represent private outdoor areas, while even-numbered values represent shared outdoor areas. When an outdoor area is not mentioned, this has a negative impact on the price. For most types of outdoor areas, the price tends to be higher for private areas than for shared ones. Private garden and private rooftop terrace have the most positive effect on price. For private balcony (7), which was by far the most common description of an outdoor area, it is unclear whether the price impact is positive or negative, as shown in Panel H of Figure 6 displaying *Outdoor Area Size_{LLM}*, which represents the number of square meters of balcony, terrace, or rooftop terrace along the x-axis. If the listed dwelling has a garden, backyard, or similar, but not a balcony, terrace, or rooftop terrace, the return value is 0. Again, very low outdoor area size has a negative impact on the price. In addition, we find a clear and positive correlation between the outdoor area size and price up to approximately 30 square meters. For some properties, we find that the outdoor area exceeds 30 square meters, but the additional positive price effect appears to diminish significantly as the outdoor areas become even larger. Once again, the model results appear economically logical, indicating that the LLM provides clear improvement.

Finally, Panel I of Figure 6 shows the interaction between the number of *Fireplaces_{LLM}* and price. As previously noted, few properties were identified as having more than one fireplace through the text analysis. The figure indicates that not having a fireplace has a negative effect on price, while an existing fireplace positively affects property prices. A fireplace may hold economic value both as a source of comfort and as an alternative means of heating.

5. Concluding Remarks

In this article, we explored the potential for enhancing automated valuation models (AVMs) by incorporating features extracted using large language models (LLMs). Our aim was to develop a pipeline to extract quantitative features from housing advertisements with GPT models. By utilizing a dataset containing 9,842 residential housing transactions from Oslo, and we constructed nine new features (housing standard, type of parking, kitchen age, bathroom age, storage size, number of storage units, type of outdoor area, size of outdoor area, and number of fireplaces) to explain property prices based on the listing text using LLM. These nine LLM features were analyzed by running a horse race among four differently specified AVMs to predict property prices. Therefore, we developed two linear (OLS) and two nonlinear (XGBoost) models, and distinguish for each of the two categories those with and without LLM features.

For the nonlinear XGBoost model, the RMSE declined from 10.53 to 7.97%, and the MAPE reduced from 6.36 to 5.39% after including the LLM features, representing an improvement of 24.31 and 15.25%, respectively. Such a strong improvement on already strong results suggest that the LLM variables contain valuable information for property valuation. If the sole purpose is to improve model performance, one can, as in Oust et al. (2020) and Birkeland et al. (2021), use information from previous transaction prices to enhance AVM estimates for property characteristics that do not change over time, but only for properties where past sales data are available.

To expand this objective and to investigate the LLM features further, their performances were plotted as SHAP dependence plots and were compared to the actual housing prices. This technique allowed us to study the price relationships between the LLM features and housing prices in detail. The price relationships from the LLM features aligned with the expectations. For example, better housing standard and having a fireplace showed a clear positive relationship with price, while the age of bathrooms and kitchens had a negative price impact. However, the method is even more detailed, as it allowed us to showcase nonlinear relationships between the variables as well as to compare, for example, different types of parking and different types of outdoor areas against each other.

The findings indicate that LLMs can effectively extract valuable information from the textual components of housing advertisements, which, when combined with traditional data, provide a more nuanced and accurate valuation model. This approach not only improves AVM accuracy but also has broader implications for real estate research, offering new avenues for exploring market dynamics that were previously difficult to quantify due to the lack of structured data. Overall, this research contributes to the growing body of literature on the ML application in real estate, particularly in the integration of LLMs for feature extraction. The results reported in this article underscore the value of combining structured and unstructured data to achieve more accurate and reliable property valuations, highlighting the promising future of AVMs in both practical and research contexts. Rather than introducing a broad methodological innovation, this study provides a focused and modular demonstration of how LLMs can be operationally integrated into an existing AVM pipeline.

The practical implications of this research are far-reaching. Enhanced AVMs can be adopted by tax authorities, banks, real estate agents, and algorithmic traders to perform more precise property valuations. From a research perspective, our study also provides a

clear and replicable pipeline for using LLMs to enrich structured housing data with text-derived features, demonstrating a valuable approach for future studies seeking to integrate unstructured information into valuation models. Moreover, the methodology employed in this study could serve as a foundation for further research into other unstructured data sources, potentially leading to even greater advancements in AVM technology.

Disclosure Statement

No potential conflict of interest was reported by the authors.

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Notes

1. The superiority of nonlinear AVMs is demonstrated by Wang et al. (2014), Kok et al. (2017), Huang (2019), Mayer et al. (2019), Bogin and Shui (2020), Su et al. (2021), Stang et al. (2023), Mamre and Sommervoll (2024).
2. See, e.g., Oust et al. (2020), Fout et al. (2022), Oust et al. (2023), Calainho et al. (2024), Mamre and Sommervoll (2024).
3. See, e.g., Yacim and Boshoff (2018), Mayer et al. (2019), Pérez-Rave et al. (2019), Bogin and Shui (2020), Pace and Hayunga (2020), Birkeland et al. (2021), Simlai (2021), Ho et al. (2021), Hjort et al. (2022); Sing et al. (2022), Yoshida et al. (2024).
4. The idea of using computers to mimic human language using natural language processing (NLP) has a long history. In the 1960s, Joseph Weizenbaum developed the first chatbot called ELIZA (Weizenbaum, 1976), which was a simple simulated conversation program between human and machine (Weizenbaum, 1966).
5. Translated into English from the original language (Norwegian).
6. We use transaction prices both to train and to evaluate the model. Both practices are fully in line with the AVM literature and with what is done in practice. However, there is a distinction between transaction price and market value, as market value is a concept referring to what one believes the property would sell for in a competitive, open market at a given point in time. Using transaction prices instead of market value may slightly penalize model performance, as market value does not reflect, for instance, the idiosyncratic dynamics of individual negotiations or auction-like settings.
7. Oust et al. (2023) studied the price effects of varying technical conditions of dwellings in the Oslo area within an AVM setting.
8. Translated into English from the original Norwegian.

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Appendix

A Model Behavior for LLM

The following is the default model behavior for the LLM. It is constructed as one string, including a set of rules for each feature the model extracts. For simplicity, it is presented in sections here. In some cases, the model is unable to return a value in the desired format. These values are set to 0.

Format⁸ Your task is to analyze the content of the real estate listing. You should return the following information about the property in the following format: 'parking': 'p_score', 'kitchen': 'k_score', 'bathroom': 'b_score', 'fireplace': 'f_score', 'storage': 's_score', 'storagesize': 'ss_score', 'outdoorarea': 'o_score', 'outdoorareaisize': 'os_score', 'dwellingstandard': 'd_score'.

The 'x_score' values are always in a numerical format and their selection is outlined below.

Parking

The 'p_score' ranges from 1 to 7, and only a single option that best matches the parking characteristics can be chosen from the list below:

1. No parking or parking not mentioned.
2. There is parking or street parking in the area, but not exclusive to the property.
3. The housing cooperative/condominium has shared parking spaces; the cooperative has multiple parking spaces.
4. The property has the option to rent a parking space.
5. The housing cooperative/condominium has parking spaces allocated based on a waiting list/seniority.
6. The property has its own outdoor parking, parking space, or carport; each unit owner has parking, and the apartment comes with an assigned parking space.
7. The property has its own garage parking.

Kitchen

The 'k_score' is equal to the calendar year (in the YYYY format) the kitchen was built or last renovated, refurbished, or upgraded, or alternatively, the year the property was built. If the year is not mentioned, return 0.

Bathroom

The 'b_score' is equal to the calendar year (in the YYYY format) the bathroom was built or last renovated, refurbished, or upgraded, or alternatively, the year the property was built. If the year is not mentioned, return 0.

Fireplaces

The 'f_score' is equal to the number of fireplaces/wood-burning stoves. If not mentioned, f score = 0. If a fireplace is mentioned but the number is not specified, assume the number is 1.

Storage Unit

The 's_score' is equal to the number of storage units available to the property. If storage is mentioned but the number of units is not specified, s_score = 1.

Storage Unit Size

The 'ss_score' is equal to the size (in square meters) of the storage unit. If there are multiple storage units, it is the total combined area of the units. If the size is not clearly stated, ss_score = 0.

Outdoor Area

The 'o_score' is set to a value in the 0 – 10 range corresponding to the description given in the following list that best matches the outdoor area characteristics (only one number can be chosen):

0. Outdoor area not mentioned
1. Private backyard
2. Shared backyard
3. Private garden
4. Shared garden
5. Private terrace
6. Shared terrace
7. Private balcony
8. Shared balcony
9. Private rooftop terrace
10. Shared rooftop terrace

Outdoor Area Size

The 'os_score' is the size (in square meters) of the balcony, terrace, or rooftop terrace. If the property has multiple outdoor areas, it is the combined area of the balconies, terraces, or rooftop terraces. If it has a garden, backyard, or similar, return 0.

Housing Standard

The d_score is set to a value in the 0 – 9 range corresponding to the description given in the following list that best matches the housing standard (only one number can be chosen):

0. Not possible to determine the standard.
1. The property requires total renovation, with serious issues and deficiencies that demand extensive work.
2. The property has a low standard and needs significant upgrades and renovations to become habitable.
3. Basic standard where the property is functional but requires modernization and minor repairs.
4. Moderate standard where the property is in decent condition but needs esthetic updates and some repairs.
5. Good standard where the property is well-maintained with some modern amenities but may need minor upgrades.
6. Very good standard where the property has several updated features, is well-maintained, and requires only minor upgrades.
7. High standard where the property is recently renovated or built with quality materials, modern amenities, and little to no need for upgrades.
8. Very high standard where the property appears luxurious with state-of-the-art amenities, exclusive materials, and high-quality details.
9. Exceptional standard where the property has unique qualities, custom solutions, advanced technology, and uses the most exclusive materials and amenities available.